

# PAPER\_285: M16 Eagle Nebula UQFF — Erosion Saturation Half-Time and $\Delta g_{\text{Max}}$

**Photoevaporation Asymptotic Saturation:  $t_{\text{half}} = \tau \cdot \ln 2 = 2.079 \text{ Myr}$**

**Classification:** UQFF 2.0 Gravitational Physics — Nebular Erosion Dynamics

**System:** M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region

**Session:** 80 | **Module:** M16\_UQFF\_MODULE.cpp (22nd C++ UQFF module)

**Author:** Daniel T. Murphy | **Date:** March 2026

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## Abstract

This paper derives the **Erosion Saturation Half-Time** ( $t_{\text{half}}$ ) and **Maximum Erosion Gravity Amplitude** ( $\Delta g_{\text{Max}}$ ) for the M16 Eagle Nebula UQFF photoevaporation term. The photoevaporation rate follows an exponential saturation  $E_{\text{rad}}(t) = E_0 \times (1 - \exp(-t/\tau))$  with e-folding time  $\tau = 3 \text{ Myr}$ . The half-erosion time  $t_{\text{half}} = \tau \cdot \ln 2 = 6.561 \times 10^{13} \text{ s} = 2.079 \text{ Myr}$  — the time at which the erosion fraction reaches half its asymptotic maximum  $E_0$ . The maximum gravity perturbation is  $\Delta g_{\text{Max}} = E_0 \times g_{\text{base}} = 4.36 \times 10^{-13} \text{ m/s}^2$ . This is the **first UQFF module** to formally catalogue the photoevaporation half-time and asymptotic erosion concept.

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## 2. Physical Background

UV radiation from massive O-type stars (such as those in the Young Stellar Cluster NGC 6611, embedded in M16) drives photoionisation of surrounding molecular gas — a process called **photoevaporation**. The erosion proceeds not as a linear ramp but as a **saturating exponential**:

$$E_{\text{rad}}(t) = E_0 (1 - e^{-t/\tau})$$

where: -  $E_0 = 0.3$  is the **asymptotic maximum fraction** (30% of mass eventually eroded)  
-  $\tau = 3 \text{ Myr}$  is the **e-folding time** (photoevaporation efficiency timescale) - The saturation arises because dense molecular cores (pillar tips) are progressively shielded by their own column density as surrounding gas is stripped

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## 3. Mathematical Derivation

### 3.1 Half-Time

The erosion half-time  $t_{\text{half}}$  is defined as the time when  $E_{\text{rad}} = E_0/2$ :

$$E_0 (1 - e^{-t_{\text{half}}/\tau}) = \frac{E_0}{2}$$

$$1 - e^{-t_{\text{half}}/\tau} = \frac{1}{2}$$

$$e^{-t_{\text{half}}/\tau} = \frac{1}{2}$$

$$t_{half} = \tau \ln(2)$$

For M16:

$$t_{half} = 9.468 \times 10^{13} \text{ s} \times \ln(2) = 9.468 \times 10^{13} \times 0.6931 = 6.561 \times 10^{13} \text{ s}$$

$$t_{half} = \mathbf{2.079 \text{ Myr}}$$

### 3.2 Maximum Gravity Amplitude

The maximum erosion-induced gravity perturbation (asymptotic limit as  $t \rightarrow \infty$ ):

$$\Delta g_{Max} = E_0 \times g_{base}$$

For M16:

$$\Delta g_{Max} = 0.3 \times 1.454 \times 10^{-12} = \mathbf{4.36 \times 10^{-13} \text{ m/s}^2}$$

### 3.3 Peak Erosion Rate

The instantaneous erosion rate at  $t = 0$  (maximum rate, before saturation):

$$\left. \frac{dE_{rad}}{dt} \right|_{t=0} = \frac{E_0}{\tau}$$

The corresponding gravity change rate:

$$\left. \frac{dg_{erode}}{dt} \right|_{t=0} = \frac{E_0}{\tau} \times g_{base} = \frac{0.3}{9.468 \times 10^{13}} \times 1.454 \times 10^{-12} = 4.61 \times 10^{-27} \text{ m/s}^2/\text{s}$$


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## 4. Saturation Profile

| Time               | t (s)                  | E_rad / E <sub>0</sub> | E_rad | g_erode (m/s <sup>2</sup> )  |
|--------------------|------------------------|------------------------|-------|------------------------------|
| 0 Myr              | 0                      | 0%                     | 0     | 0                            |
| t_half = 2.079 Myr | 6.561×10 <sup>13</sup> | <b>50%</b>             | 0.150 | <b>2.18×10<sup>-13</sup></b> |
| τ = 3 Myr          | 9.468×10 <sup>13</sup> | 63.2%                  | 0.190 | 2.76×10 <sup>-13</sup>       |
| 5 Myr              | 1.578×10 <sup>14</sup> | 81.1%                  | 0.243 | 3.54×10 <sup>-13</sup>       |
| ∞ (asymptote)      | → ∞                    | <b>100%</b>            | 0.300 | <b>4.36×10<sup>-13</sup></b> |

**Key insight:** At  $\tau = 3$  Myr (the e-folding time), erosion has consumed only 63.2% of its capacity, NOT 100%. Half-erosion occurs earlier at 2.079 Myr. The pillar structure of M16 means the ~5700 ly “Pillars of Creation” are still observed today because erosion saturates — it cannot fully strip the densest pillar cores within observable timescales.

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## 5. UQFF 2.0 Context

In the full M16  $g_{\text{total}}$  equation, the erosion half-time governs the **temporal shape of  $g_{\text{dyn}}(t)$** :

$$g_{\text{dyn}}(t) = g_{\text{base}} \times (1 + M_{\text{sf}}) \times (1 - E_0(1 - e^{-t/\tau}))$$

The transition from rapid to slow erosion occurs at  $t_{\text{half}} = 2.079$  Myr. For the UQFF simulation ( $t$  stepping from 0 to  $t_{\text{max}}$ ), the half-time provides a natural **inflection point** in the dynamic gravity trajectory — before  $t_{\text{half}}$ , erosion is dominant; after  $t_{\text{half}}$ , star formation accumulation dominates (since  $M_{\text{sf}}$  grows linearly while  $E_{\text{rad}}$  asymptotes).

### Crossover Time

The era when SFR growth exactly compensates erosion ( $d\Phi_{\text{dm}}/dt = 0$  — maximum  $\Phi_{\text{dm}}$ ):

$$\frac{d\Phi_{\text{dm}}}{dt} = \text{SFR}_{\text{rate}} \times (1 - E_{\text{rad}}) - (1 + M_{\text{sf}}) \times \frac{E_0}{\tau} e^{-t/\tau} = 0$$

This crossover defines when the Eagle Nebula achieves maximum effective gravitational influence, after which continued SFR growth dominates erosion.

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## 6. Wolfram KB Term

M16UQFF: $t_{\text{half}}=\tau*\text{Log}[2]=6.561\text{e}13\text{s}=2.079\text{Myr}$ ;       $\text{DeltaGMax}=E_0*g_{\text{base}}=4.36\text{e}-13 \text{ m/s}^2$  [PAPER\_285]

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## 7. Cross-References

- **PAPER\_284**: Dual Mass Co-Action Product ( $\Phi_{\text{dm}} = (1+M_{\text{sf}}) \times (1-E_{\text{rad}})$ )
  - **PAPER\_286**: Nebular Friedmann Redshift ( $\kappa_{\text{neb}}$ ,  $z=0.0015$ )
  - **M16\_UQFF\_MODULE.cpp**: Full UQFF 2.0 C++ implementation (22nd module)
  - **CondensedPhysics3.py**: M16ErosionSaturationHalfTimeCalculator
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